

CHAPTER 2: CIRCUIT THEORY PRIMER

①

Electronics is all about controlling electrons.

* Charge (Q): quantity of electrons. Coulombs (C)

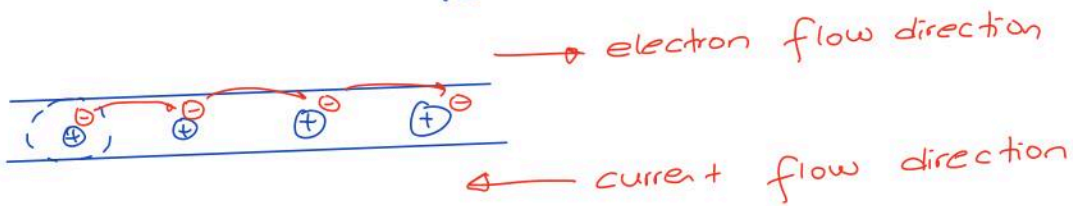
$$1 e^- \rightarrow -1.602 \times 10^{-19} \text{ C}$$

$$-1 \text{ C} \rightarrow 6.241 \times 10^{18} \text{ electrons}$$

* Current (I): flow rate of charged particles
1 C of charge moves across a cross-section
in 1 second $\rightarrow$ 1 Ampere current

Current direction is the same as the flow direction
of positive charges.

$$1 \text{ A} = \frac{1 \text{ C}}{1 \text{ s}}$$



* Voltage (V): electrical potential difference between
two points in a circuit
 $\uparrow$
always relative

* amount of desire for charges to move from
one place to the other.

* 1V can deliver 1J of energy to 1C of charge:

$$E = Q \cdot V \quad \Rightarrow \quad \text{Volts} = \frac{\text{Joules}}{\text{Coulomb}}$$

* Resistance: degree to which a conductor opposes
current flow through it. Ohms (Ω)

Good conductors: copper, gold, silver, ...

Bad conductors (insulators): wood, plastic, glass,
ceramics, ...

* Instantaneous power: $P(t) = v(t) \cdot i(t)$

Water Flow Analogy

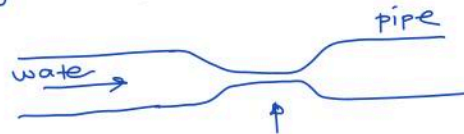
(2)

charge (C) $\longleftrightarrow$ volume of water (liters)

current $(A = C/s)$ $\longleftrightarrow$ flow rate of water (liters/sec)

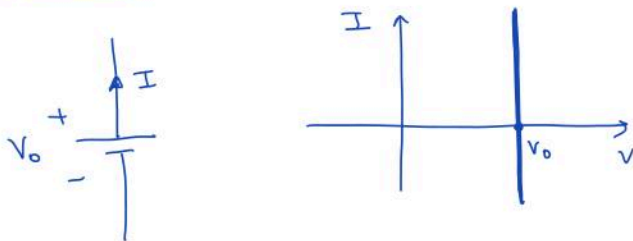
voltage (V) $\longleftrightarrow$ Pressure difference between two points in a hydraulic system

resistance (Ω) $\longleftrightarrow$ friction/constriction in a pipe system

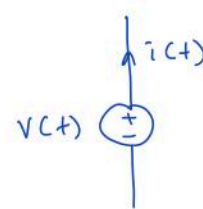


Energy Sources:

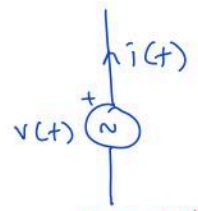
a) Voltage Source:



DC voltage source
(no time variation)
DC: stands for "direct current"



general voltage source
(could be time-varying)



AC voltage source

AC: stands for "alternating current"

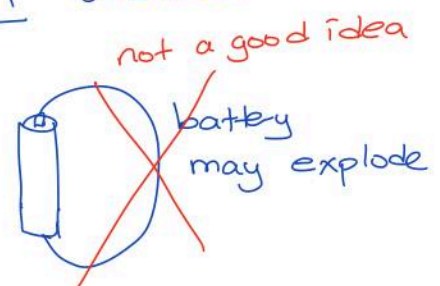
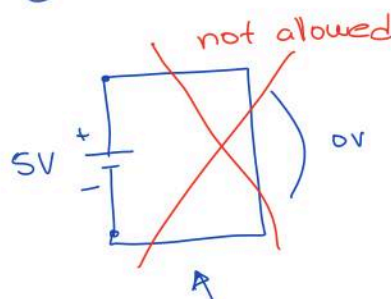
* a voltage source can supply any current to keep the voltage at the specified value.

* example: batteries

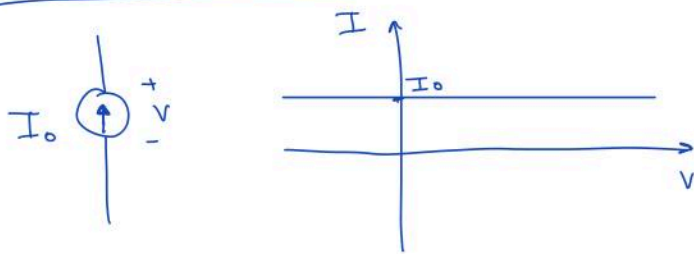
* $P = V \cdot I \Rightarrow$ ideal voltage source can supply infinite power

* Shorting a voltage source is NOT allowed.

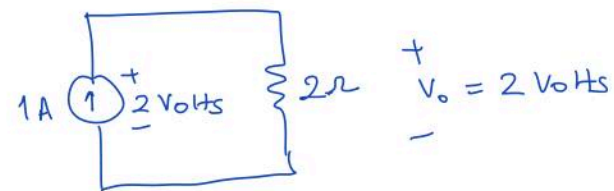
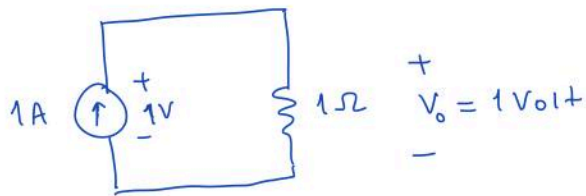
↑
short circuiting



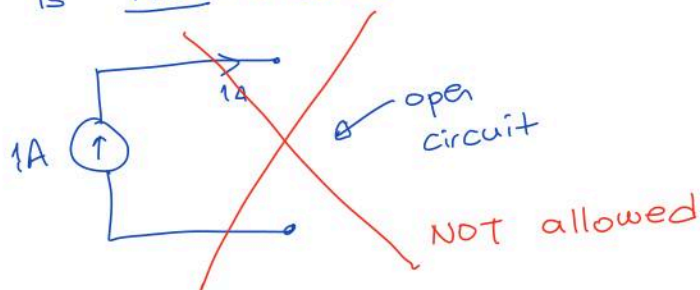
b) Current Source :



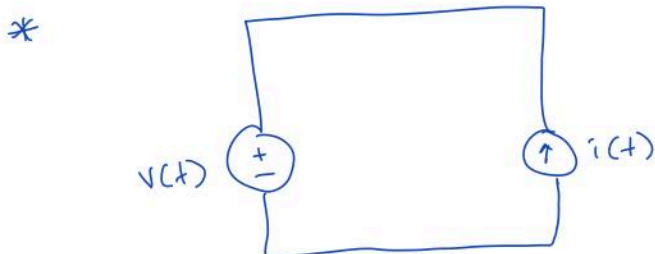
* current source needs to supply the specified current. So, it can supply any voltage to keep the current at the specified value.



* So, leaving the leads of a current source open is NOT allowed.
↑
open circuited



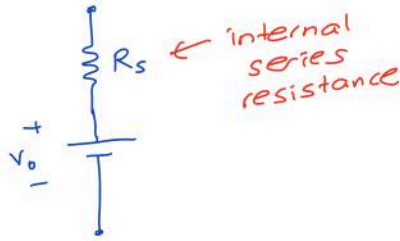
* example : battery charger



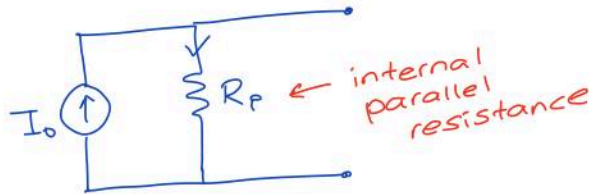
What happens?
 Is this allowed?
 (to be answered later)

* These are ideal current/voltage sources.
In real life, there are internal resistances.

Realistic
voltage
source :



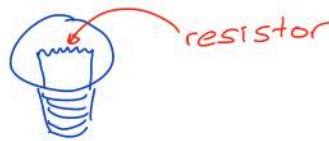
Realistic
current
source :



Resistors :

- dissipate energy
- convert electrical energy into heat energy

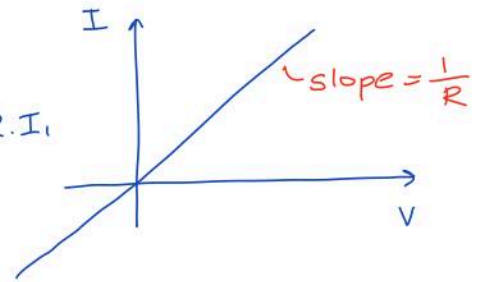
example : light bulb



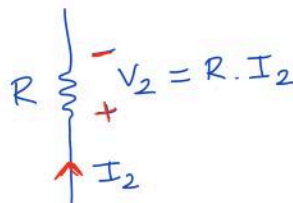
Ohm's Law : voltage across a resistor is proportional to the current that passes through it.

$$R = \frac{V}{I}$$

resistance, in Ohms (Ω)



OR



if same resistor
in the same circuit:

$$I_1 = -I_2$$

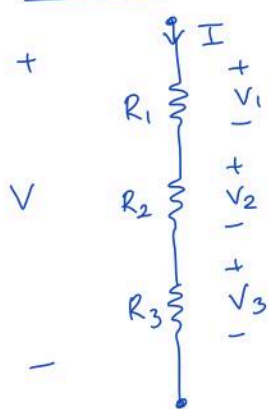
$$V_1 = -V_2$$

* Inverse of R is called conductance

$$G = \frac{1}{R} = \frac{I}{V}$$

conductance, in units of Siemens (S)
or mho

Resistors in Series: Current is the same, voltage is different (5)



$$R_T = \frac{V}{I} = \frac{V_1 + V_2 + V_3}{I}$$

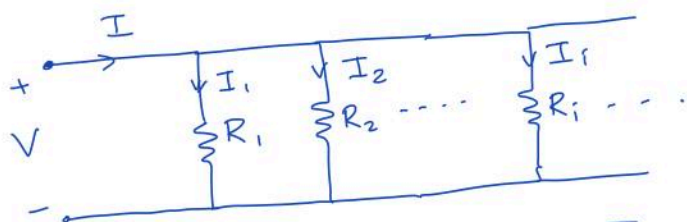
total resistance

$$= \frac{R_1 I + R_2 I + R_3 I}{I}$$

$$= R_1 + R_2 + R_3$$

In general, $R_T = \underbrace{\sum_i R_i}_{\text{increased resistance}}$ for series connection

Resistors in Parallel: voltage is the same, current is different



Total current: $I = \sum_i I_i$

$$R_T = \frac{V}{I}$$

$$= \frac{V}{(I_1 + I_2 + \dots)} = \frac{V}{\left(\frac{V}{R_1} + \frac{V}{R_2} + \dots\right)}$$

$$= \left(\frac{1}{R_1} + \frac{1}{R_2} + \dots\right)^{-1}$$

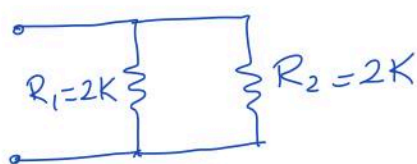
$$R_T = \left(\sum_i \frac{1}{R_i}\right)^{-1}$$

decreased resistance

⇒ more path for the current to flow

↙ denotes parallel connection

Example:



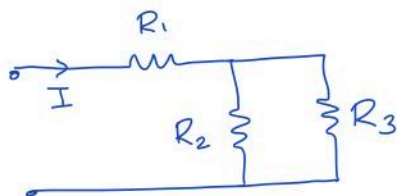
$$R_T = R_1 \parallel R_2$$

$$= \left(\frac{1}{R_1} + \frac{1}{R_2}\right)^{-1} = \frac{R_1 R_2}{R_1 + R_2}$$

$$= \frac{2K \cdot 2K}{2K + 2K} = \frac{4 \cdot 10^6}{4 \cdot 10^3} = 1K\Omega$$

↑ decreased resistance

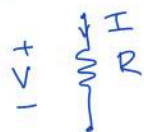
Example :



$$R_T = (R_2 // R_3) + R_1$$

$$= \frac{R_2 R_3}{R_2 + R_3} + R_1$$

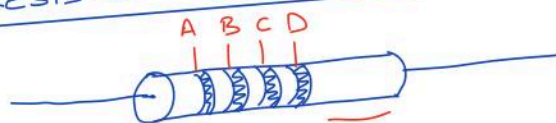
Power Delivered to a resistor :



$$P = V \cdot I = (I \cdot R) \cdot I = I^2 R$$

$$= V \cdot \left(\frac{V}{R}\right) = \frac{V^2}{R}$$

Resistor color codes :



A: 1st significant figure of resistance

B: 2nd " " "

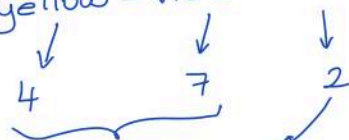
C: Multiplier

D: Tolerance (accuracy of reported resistance value)

gold 5%
silver 10%
no color 20%

example :

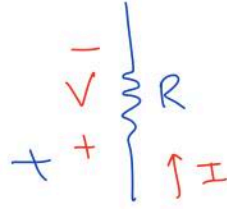
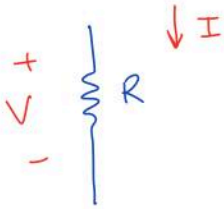
yellow - violet - red (for A-B-C)



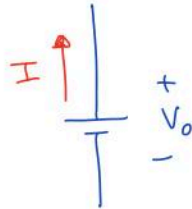
$$47 \times 10^2 = 4700 \Omega = 4.7 \text{ k}\Omega$$

Analysis of Electrical Circuits:

* Reference directions for current and voltage: (VERY IMPORTANT)

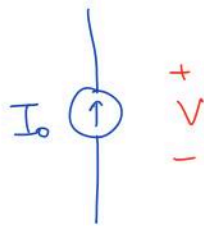


Voltage Source:



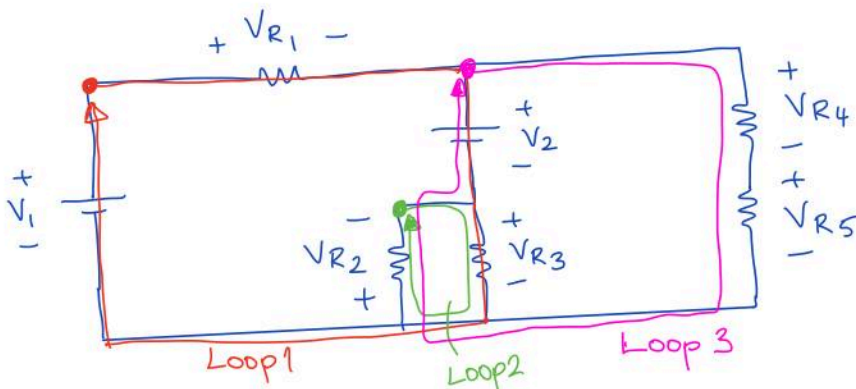
(starts from positive lead, when the voltage source is delivering power)

Current Source:



Kirchhoff's Voltage Law (KVL): The sum of voltages around any loop is zero.

* Pay attention to directions.



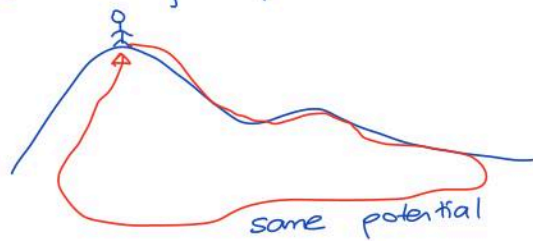
$$\text{Loop 1: } V_{R1} + V_2 + V_{R3} - V_1 = 0$$

$$\text{Loop 2: } V_{R3} + V_{R2} = 0$$

$$\text{Loop 3: } V_{R4} + V_{R5} + V_{R2} - V_2 = 0$$

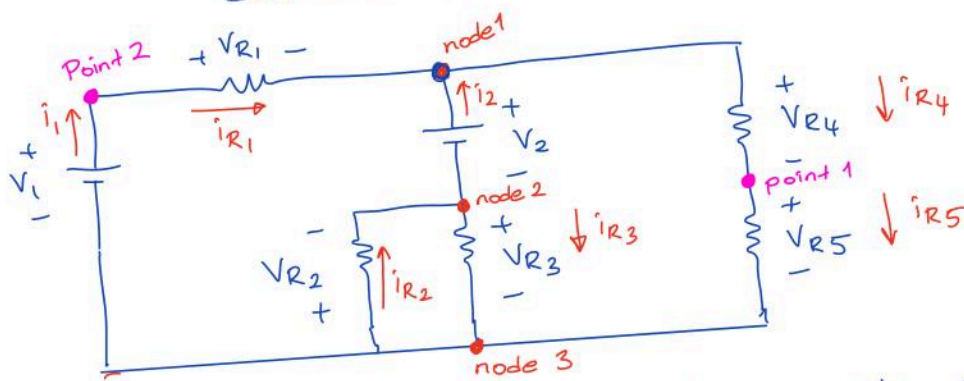
Conservation of potential

8



Kirchhoff's Current Law (KCL): The sum of currents flowing into a node is equal to the sum of the currents flowing out of a node.

Node: a point in the circuit where more than two elements are connected together.



* make sure directions of currents and voltages are consistent.

Node 1: $i_{R1} + i_2 = i_{R4}$

Node 2: $i_{R2} = i_2 + i_{R3}$

Node 3: $i_{R3} + i_{R5} = i_1 + i_{R2}$

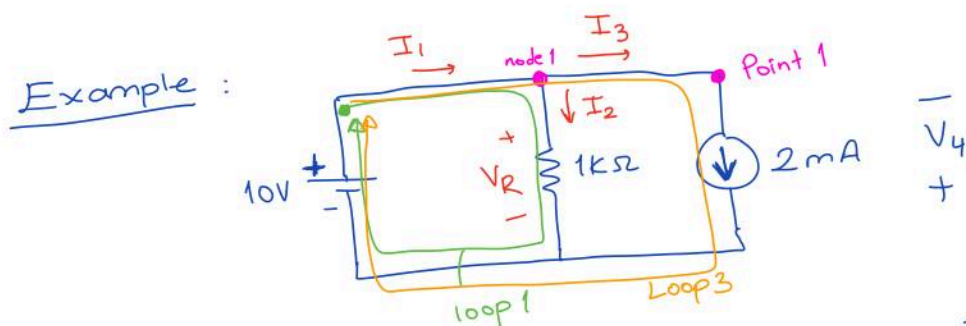
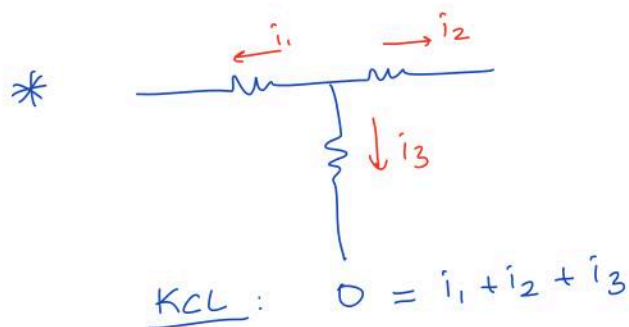
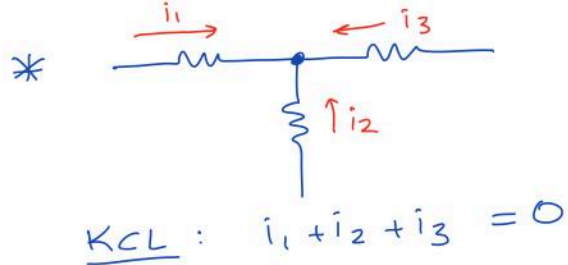
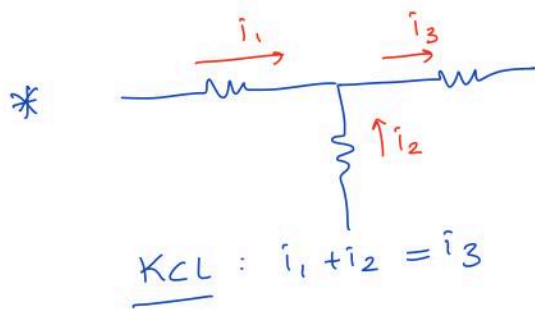
Also,

Point 1: $i_{R4} = i_{R5}$

Point 2: $i_1 = i_{R1}$

} so, no need to assign different currents to elements on the same branch.

* Conservation of charge (or mass)
incoming flow = outgoing flow.



KVL at Loop 1 : $V_R - 10V = 0 \Rightarrow V_R = 10V$

I_2 : Apply Ohm's Law

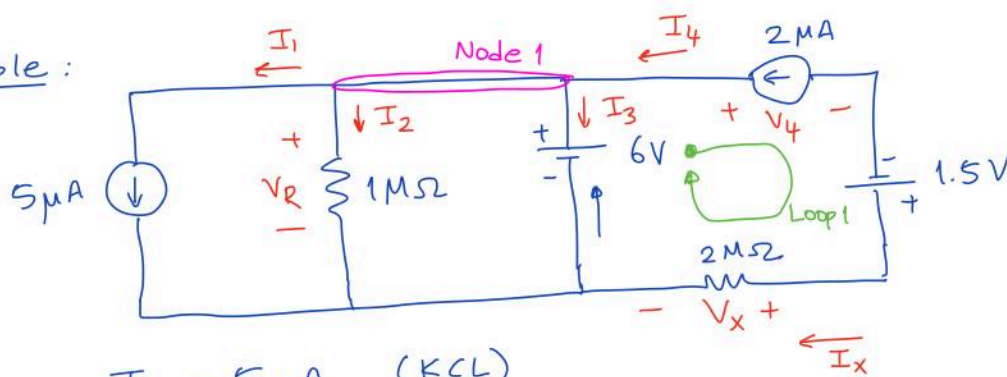
$$I_2 = \frac{V_R}{1k\Omega} = \frac{10}{1k\Omega} = 10 \text{ mA}$$

KCL at Point 1 : $I_3 = 2 \text{ mA}$

KCL at node 1 : $I_1 = I_2 + I_3 = 10 \text{ mA} + 2 \text{ mA} = 12 \text{ mA}$

KVL at Loop 3 : $-V_4 - 10V = 0 \Rightarrow V_4 = -10V$

Example:



$$I_1 = 5\mu A \quad (\text{KCL})$$

$$I_4 = 2\mu A \quad (\text{KCL})$$

$$V_R = 6V \quad (\text{KVL})$$

$$I_2 = \frac{V_R}{1M\Omega} = \frac{6V}{1M\Omega} = 6\mu A \quad (\text{Ohm's Law})$$

KCL at Node 1:

$$I_4 = I_1 + I_2 + I_3$$

$$2\mu A = 5\mu A + 6\mu A + I_3$$

$$I_3 = -9\mu A$$

$$I_x = -I_4 = -2\mu A$$

$$V_x = I_x \cdot 2M\Omega = -2\mu A \cdot 2M\Omega = -4V$$

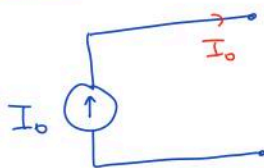
KVL at Loop 1:

$$V_4 - 1.5V + V_x - 6V = 0$$

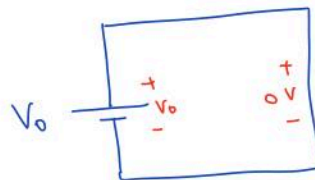
$$V_4 - 1.5 - 4 - 6 = 0$$

$$V_4 = 11.5V$$

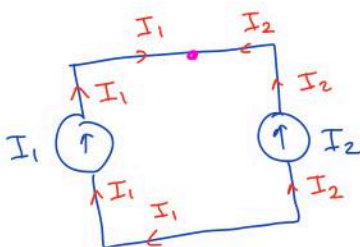
Is this circuit allowed?



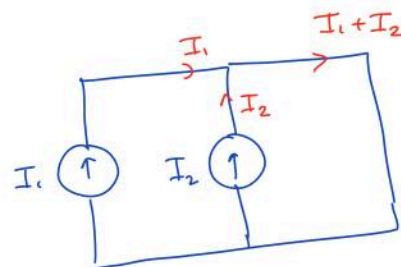
✗
Violates KCL
(unless $I_o = 0$)



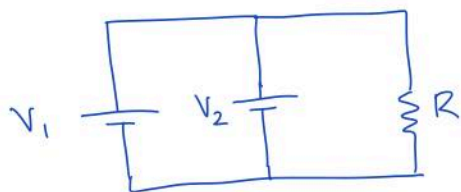
✗
Violates KVL
(unless $V_0 = 0$)



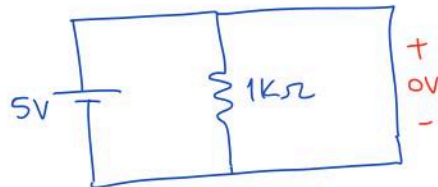
✗
Violates KCL
unless $I_1 = -I_2$



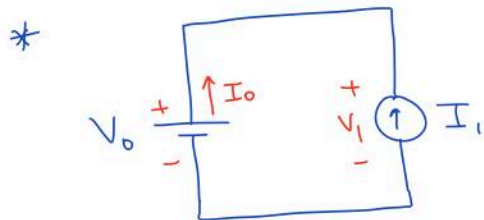
✓



X
Not allowed
unless $V_1 = V_2$
(Violates KVL)



X
Not allowed
(violates KVL)



✓ allowed

$$I_0 = -I_1$$

$$V_1 = V_0$$

$$P = I \cdot V$$

for current source :

$$P_i = I_1 \cdot V_1 = I_1 \cdot V_0$$

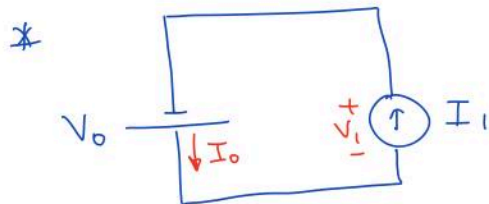
($P_i > 0 \Rightarrow$ delivering power
if $I_1 > 0, V_0 > 0$)

for voltage source :

$$P_o = I_0 \cdot V_0 = -I_1 \cdot V_0$$

($P_o < 0 \Rightarrow$ receiving power
if $I_1 > 0, V_0 > 0$)

* This is like charging a battery using a current source.



✓ allowed

$$I_0 = I_1$$

$$V_1 = -V_0$$

for current source :

$$P_i = I_1 \cdot V_1 = -I_1 \cdot V_0$$

($P_i < 0 \Rightarrow$ receiving power)

for voltage source :

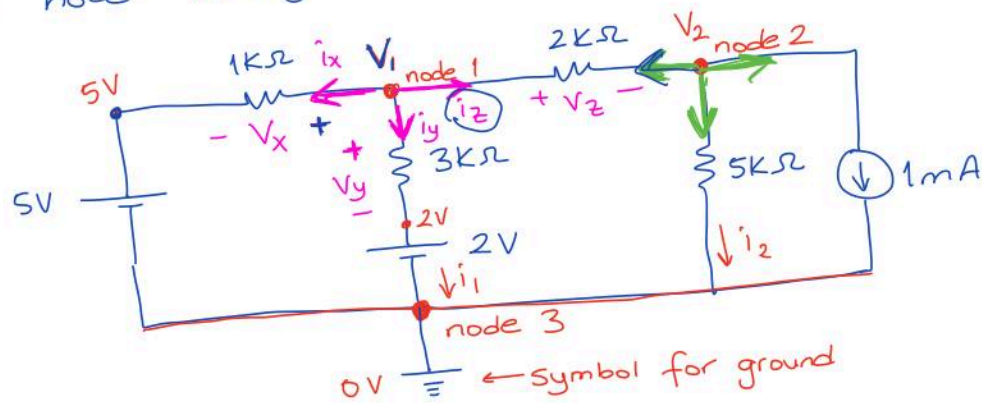
$$P_o = I_0 \cdot V_0 = I_1 \cdot V_0$$

($P_o > 0 \Rightarrow$ delivering power)

Node Analysis : Most preferred method to analyze circuits (12)

- 1) Select a common node $\Rightarrow$ ground node. We assign zero reference voltage to this node.
- 2) Define node voltages with respect to the common node.
- 3) Write KCL at each node, expressing branch currents in terms of node voltages and sources.
Everything should be written in terms of node voltages or sources.
- 4) Solve these equations simultaneously to find node voltages.
- 5) You may find the requested voltages or currents from the node voltages.

Example :



$$i_1 = ?$$
$$i_2 = ?$$

KCL at Node 1 : $i_x + i_y + i_z = 0$

$$i_x = \frac{V_x}{1k\Omega} = \frac{V_1 - 5V}{1k\Omega}$$

$$i_y = \frac{V_y}{3k\Omega} = \frac{V_1 - 2V}{3k\Omega}$$

$$i_z = \frac{V_z}{2k\Omega} = \frac{V_1 - V_2}{2k\Omega}$$

$$\frac{V_1 - 5}{1k\Omega} + \frac{V_1 - 2}{3k\Omega} + \frac{V_1 - V_2}{2k\Omega} = 0$$

KCL at Node 2 : $\frac{V_2 - V_1}{2k\Omega} + \frac{V_2 - 0}{5k\Omega} + 1mA = 0$

2 Equations
2 Unknowns

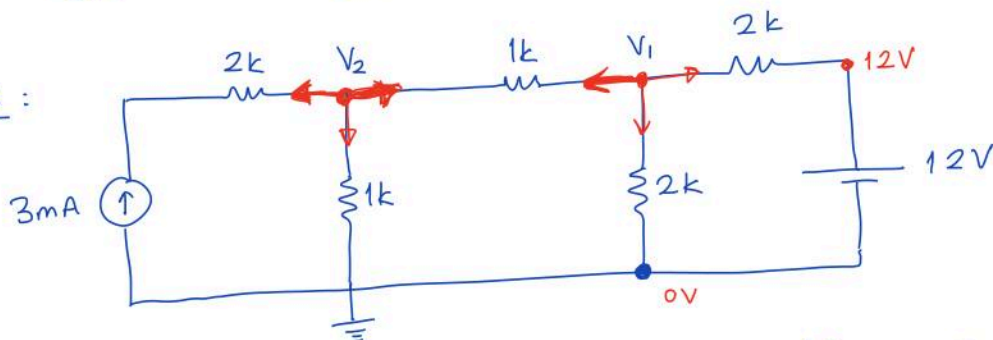
Solve simultaneously :

$$V_1 = 3.35V$$
$$V_2 = 0.95V$$

$$i_1 = \frac{V_1 - 2}{3k} = \frac{3.35 - 2}{3k\Omega} = 0.45 \text{ mA}$$

$$i_2 = \frac{V_2 - 0}{5k} = \frac{0.95}{5k} = 0.19 \text{ mA}$$

Example :



KCL at Node 1 : $\frac{V_1 - V_2}{1k} + \frac{V_1 - 0}{2k} + \frac{V_1 - 12}{2k} = 0$
(2)

$$4V_1 - 2V_2 = 12$$

KCL at Node 2 : $-3\text{mA} + \frac{V_2 - 0}{1k} + \frac{V_2 - V_1}{1k} = 0$
(1k)

$$2V_2 - V_1 = 3$$

$$+ 4V_1 - 2V_2 = 12$$

$$\frac{+}{3V_1 = 15} \Rightarrow$$

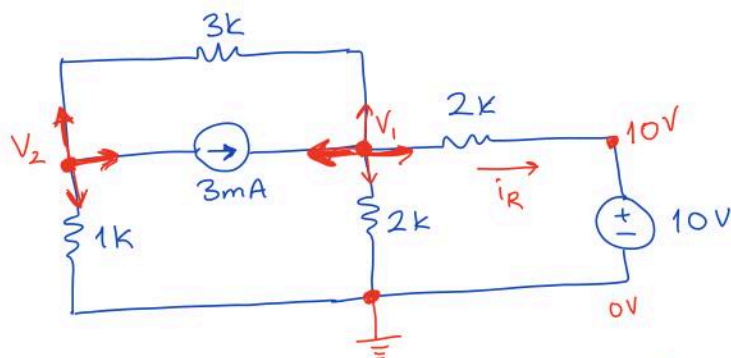
$$\boxed{V_1 = 5V}$$

$$2V_2 - 5 = 3$$

$$\Rightarrow$$

$$\boxed{V_2 = 4V}$$

Example :



$i_R = ?$

KCL at Node 1 : $-3\text{mA} + \frac{V_1}{2k} + \frac{V_1 - V_2}{3k} + \frac{V_1 - 10}{2k} = 0$
(6k) (3) (2) (3)

$$8V_1 - 2V_2 = 48$$

$$4V_1 - V_2 = 24 \dots \dots (\text{Eqn. 1})$$

KCL at Node 2: $3\text{mA} + \frac{V_2 - 0}{1\text{k}} + \frac{V_2 - V_1}{3\text{k}} = 0$

(14)

$$4V_2 - V_1 = -9 \quad \dots (\text{Eqn. 2})$$

$$16V_1 - 4V_2 = 96 \quad \dots (\text{Eqn. 1}) \times 4$$

+

$$15V_1 = 87$$

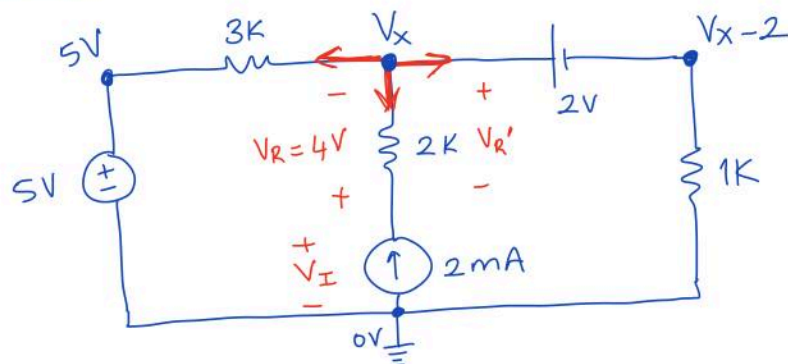
$$V_1 = \frac{87}{15} = \boxed{5.8\text{V}}$$

$$4V_2 - 5.8 = -9$$

$$V_2 = \frac{-9 + 5.8}{4} = \frac{-3.2}{4} = \boxed{-0.8\text{V}}$$

Finally, $i_R = \frac{V_1 - 10}{2\text{k}} = \frac{5.8 - 10}{2\text{k}} = \boxed{-2.1\text{mA}}$

Example: Nodal analysis: A common mistake



Do not ignore the voltage of current source

$$V_I \neq 0$$

KCL: $\frac{V_x - 5}{3\text{k}} - 2\text{mA} + \frac{V_x - 2}{1\text{k}} = 0 \Rightarrow 4V_x = 17$

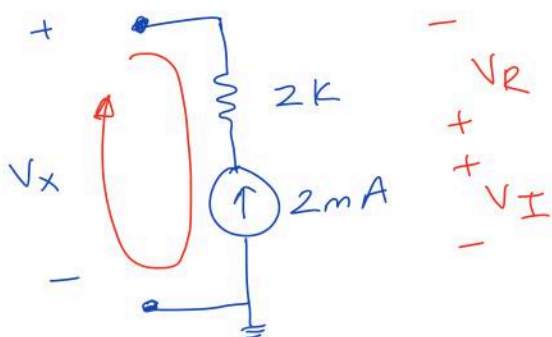
$$\boxed{V_x = 4.25\text{V}}$$

$$V_x = -V_R + V_I$$

$$4.25 = -4 + V_I$$

$$\Rightarrow \boxed{V_I = 8.25\text{V}} \quad (\text{not zero!})$$

Also, $V_x = V_{R'} + V_I = -V_R + V_I$ (same as before)



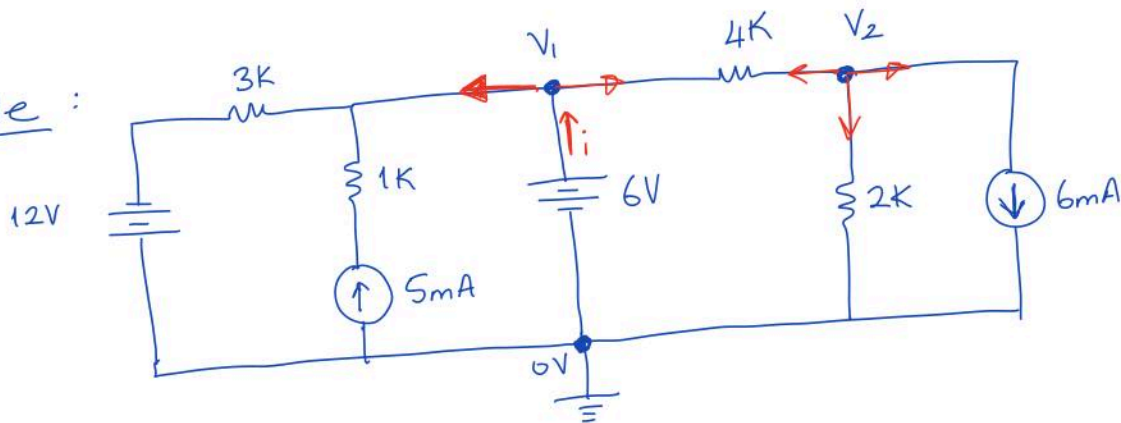
KVL:

$$-V_R + V_I - V_x = 0$$

$$V_x = -V_R + V_I$$

(same as before)

Example :

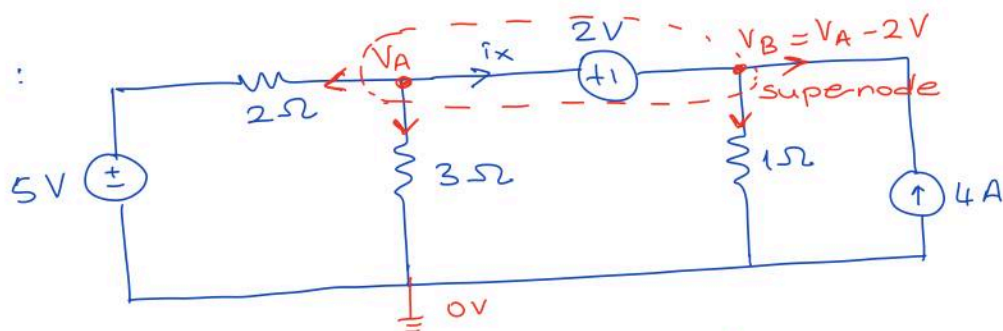


$V_1 = 6V$ (no need for KCL)
 KCL at Node 2: $\frac{V_2 - 6}{4K} + \frac{V_2}{2K} + 6mA = 0$

$$3V_2 + 18 = 0$$

$$V_2 = -6V$$

Example :



$i_x = ?$

$$V_B = V_A - 2$$

KCL at $V_A + V_B$: $\frac{V_A - 5}{2} + \frac{V_A}{3} + \frac{V_B}{1} - 4 = 0$

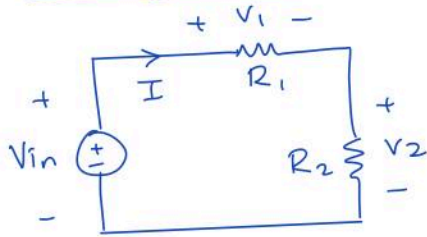
$$11V_A = 15 + 36 = 51$$

$$V_A = \frac{51}{11} V \approx 4.64V$$

$$V_B = V_A - 2 = 2.64V$$

Then, $i_x = \frac{V_B}{1} - 4 = 2.64 - 4 = -1.36A$

Voltage Divider:

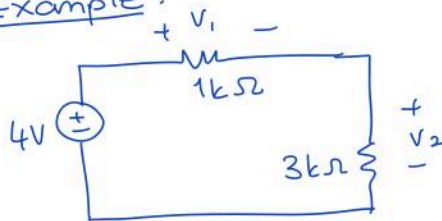


$$I = \frac{V_{in}}{R_1 + R_2}$$

$$V_1 = R_1 \cdot I = V_{in} \cdot \underbrace{\frac{R_1}{R_1 + R_2}}_{\text{voltage divider term}}$$

$$V_2 = R_2 \cdot I = V_{in} \cdot \underbrace{\frac{R_2}{R_1 + R_2}}_{\text{voltage divider term}}$$

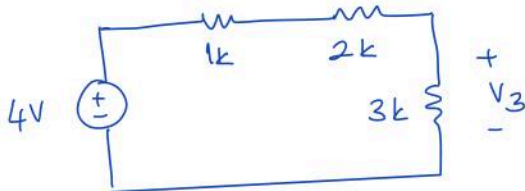
Example:



$$V_1 = 4 \cdot \frac{1k}{1k + 3k} = 1V$$

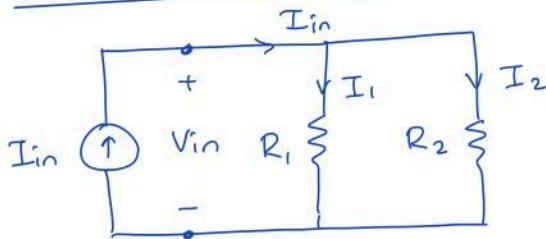
$$V_2 = 4 \cdot \frac{3k}{1k + 3k} = 3V$$

Example:



$$V_3 = 4 \cdot \frac{3k}{1k + 2k + 3k} = 2V$$

Current Divider:



$$I_{in} = \frac{V_{in}}{R_1 \parallel R_2}$$

$$I_1 = \frac{V_{in}}{R_1}$$

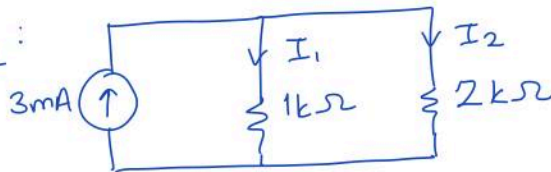
$$\left. \begin{array}{l} I_{in} = \frac{V_{in}}{R_1 \parallel R_2} \\ I_1 = \frac{V_{in}}{R_1} \end{array} \right\} \frac{I_{in}}{I_1} = \frac{R_1}{R_1 \parallel R_2}$$

$$I_1 = I_{in} \cdot \frac{\cancel{R_1} \cdot R_2}{\cancel{R_1} + R_2}$$

$$I_1 = I_{in} \cdot \underbrace{\frac{R_2}{R_1 + R_2}}_{\text{current divider term}}$$

Similarly, $I_2 = I_{in} \cdot \underbrace{\frac{R_1}{R_1 + R_2}}_{\text{current divider term}}$

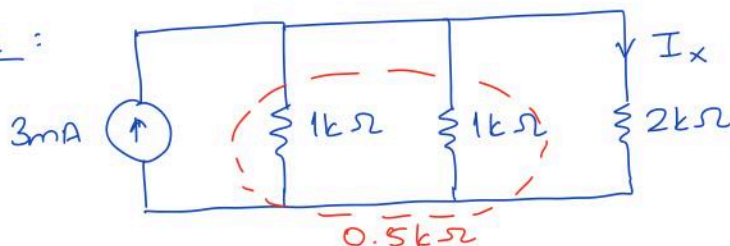
Example:



$$I_1 = 3mA \cdot \frac{2k}{1k + 2k} = 2mA$$

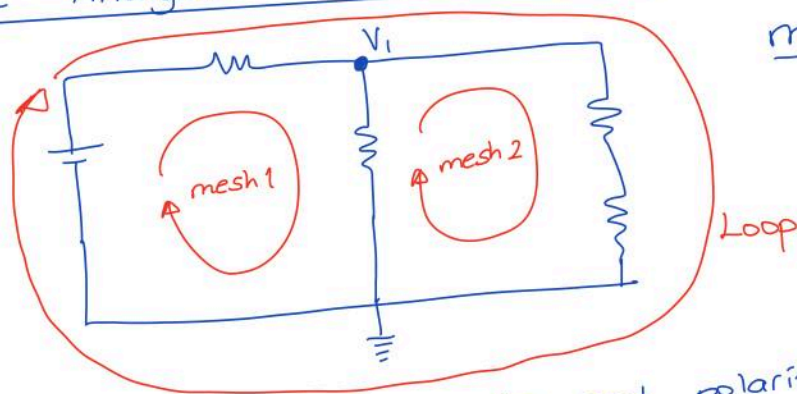
$$I_2 = 3mA \cdot \frac{1k}{1k + 2k} = 1mA$$

Example:



$$I_x = 3mA \cdot \frac{0.5k}{0.5k + 2k} = 0.6mA$$

Node Analysis vs. Mesh Analysis



mesh: a loop that does not contain any other loops

- assign mesh currents and polarities
- apply KVL, expressing voltages in terms of mesh currents.
- * Mesh currents cannot be measured directly, whereas node voltages can be measured.
- * Typically, node analysis gives fewer number of equations.
1 node equation vs. 2 mesh equations

Power Line

In Turkey: 220 Vrms, 50 Hz

In USA: 110 Vrms, 60 Hz

Other places: 110 Vrms, 220-240 Vrms
50 Hz or 60 Hz

About Rms (Root-mean-square)

* Instantaneous power: $p(t) = v(t) i(t)$

Average power: $P = \frac{1}{T} \int_0^T v(t) i(t) dt$

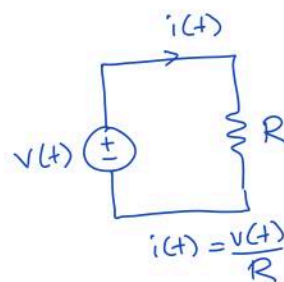
$$= \frac{1}{T} \int_0^T v(t) \cdot \frac{v(t)}{R} dt$$

$$= \frac{1}{R} \cdot \frac{1}{T} \int_0^T v^2(t) dt$$

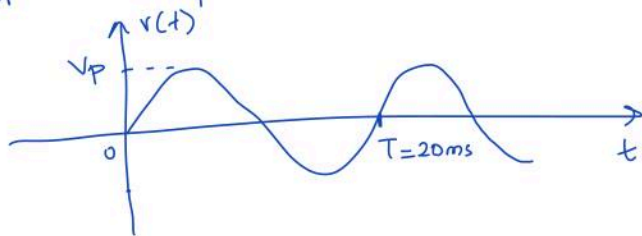
Now, define: $V_{rms} = \sqrt{\frac{1}{T} \int_0^T v^2(t) dt}$

then,

$$P = \frac{V_{rms}^2}{R} = i_{rms}^2 \cdot R$$



* For the power line:



$$v(t) = V_p \cdot \sin(\omega t)$$

In Turkey: $f = 50 \text{ Hz}$
 $\omega = 2\pi 50 \text{ rps}$
 $T = \frac{1}{f} = \frac{1}{50} = 0.02 \text{ s} = 20 \text{ ms}$

$$\begin{aligned} V_{rms} &= \sqrt{\frac{1}{T} \int_0^T v^2(t) dt} = \sqrt{\frac{1}{T} \int_0^T V_p^2 \sin^2(\omega t) dt} \\ &= \sqrt{\frac{V_p^2}{T} \int_0^T \frac{1 - \cos(2\omega t)}{2} dt} \\ &= V_p \sqrt{\frac{1}{2T} \left[\int_0^T dt - \int_0^T \cos(2\omega t) dt \right]} \\ &= V_p \cdot \sqrt{\frac{1}{2T} [T - 0]} \end{aligned}$$

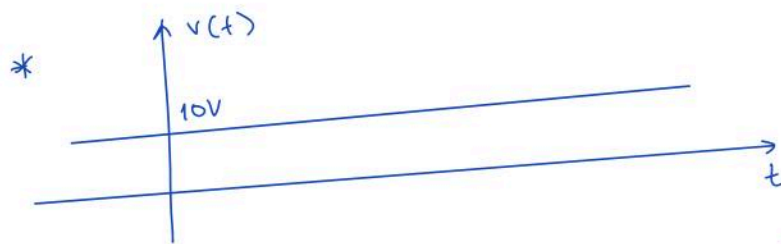
Period $T' = \frac{T}{2}$
 integrating over 2 periods
 So, this equals zero

$V_{rms} = \frac{V_p}{\sqrt{2}}$

 for sinusoidal voltage

* In Turkey, $V_{rms} = 220 \text{ Volts}$
 So, $V_p = 220 \times \sqrt{2} = 311 \text{ Volts}$

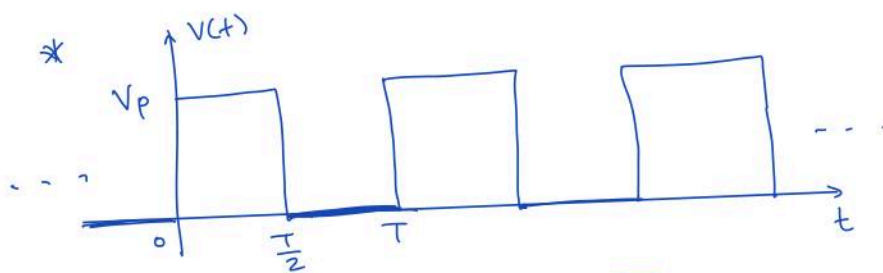
* What is the RMS value of the following?



DC voltage
 $v(t) = 10 \text{ V} = V_{DC}$
 $V_{rms} = 10 \text{ V}$

$$\begin{aligned} V_{rms} &= \sqrt{\frac{1}{T} \int_0^T v^2(t) dt} = \sqrt{\frac{1}{T} \int_0^T V_{DC}^2 dt} \\ &= \sqrt{\frac{1}{T} V_{DC}^2 \cdot T} = V_{DC} \end{aligned}$$

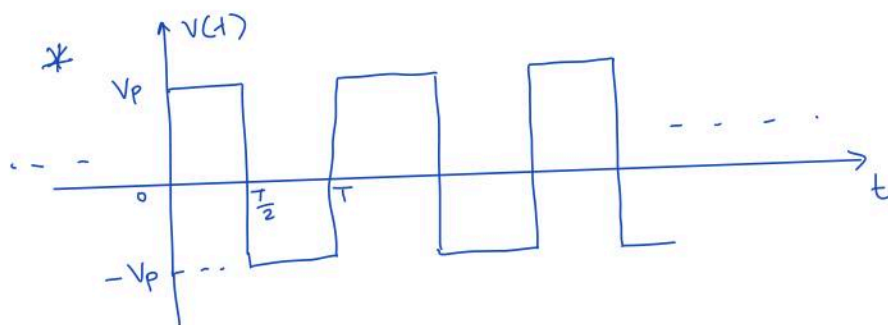
$$P_{ave} = \frac{V_{DC}^2}{R}$$



square wave

$$V_{rms} = \sqrt{\frac{1}{T} \int_0^T v^2(t) dt} = \sqrt{\frac{1}{T} \left[\int_0^{T/2} V_p^2 dt + \int_{T/2}^T 0 dt \right]}$$

$$= \sqrt{\frac{1}{T} V_p^2 \cdot \frac{T}{2}} = \frac{V_p}{\sqrt{2}}$$

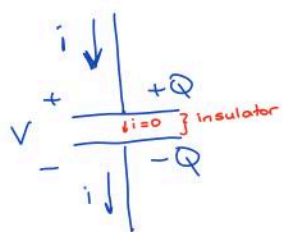


square wave

$$V_{rms} = V_p$$

Capacitors:

- Store electrical charge, Q



$$C = \frac{Q}{V}$$

charge in Coulombs
voltage in Volts

capacitance in Farads (F)

* Charge stored is proportional to the applied voltage

$$Q(t) = Q(0) + \int_0^t i(\xi) d\xi$$

Take derivative of both sides:

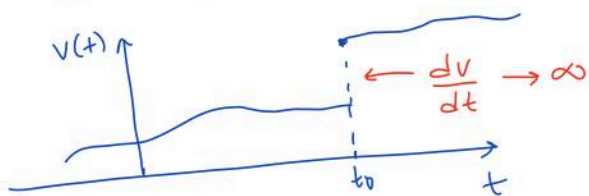
$$\frac{dQ}{dt} = i(t) \quad \rightarrow \quad Q = CV$$

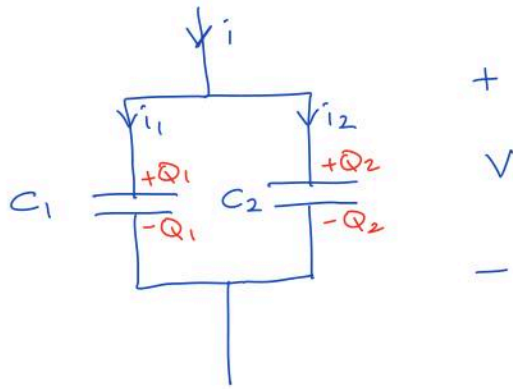
$$\frac{d(CV)}{dt} = i(t)$$

$$\boxed{C \frac{dV}{dt} = i(t)}$$

capacitors try to oppose change in voltage

*** Voltage of a capacitor cannot change instantaneously!



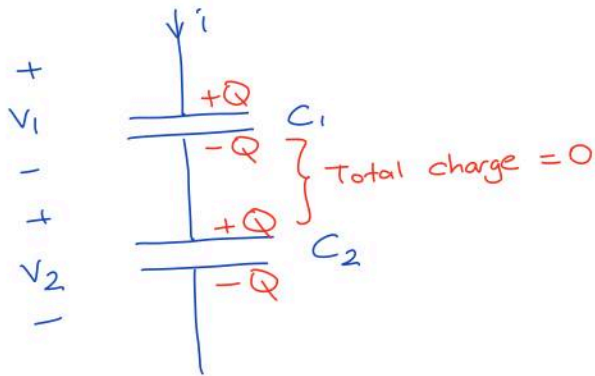
Parallel Capacitors :

Charges add

$$C_T = \frac{Q_T}{V} = \frac{Q_1 + Q_2}{V}$$

$$= \frac{C_1 V + C_2 V}{V}$$

$$C_T = C_1 + C_2$$

Serial capacitors :

voltages add

$$C_T = \frac{Q}{V_T} = \frac{Q}{V_1 + V_2}$$

$$= \frac{Q}{\frac{Q}{C_1} + \frac{Q}{C_2}}$$

$$C_T = \left(\frac{1}{C_1} + \frac{1}{C_2} \right)^{-1}$$

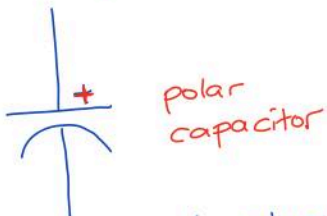
(reduced capacitance)

* Capacitors do not dissipate energy. They store energy in the form of electrical field.

* Energy stored in a capacitor:

$$E = \frac{CV^2}{2}$$

Note : Capacitors with large capacitance may have polarities.



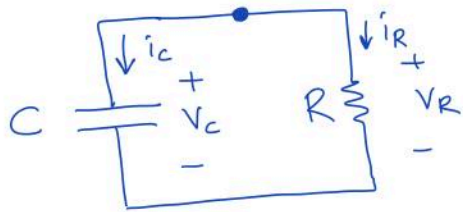
polar capacitor



non-polar capacitor

Voltage must always remain positive on the side indicated

RC Circuit :



Assume capacitor is initially charged,
with $V_C(0) = V_0$

$$i_C + i_R = 0 \quad (\text{KCL})$$

$$i_C = C \frac{dV_C}{dt}, \quad i_R = \frac{V_R}{R} = \frac{V_C}{R}$$

So,

$$C \frac{dV_C}{dt} + \frac{V_C}{R} = 0$$

$$\frac{dV_C}{dt} + \frac{1}{RC} V_C = 0$$

} 1st - order
ordinary differential equation

Then the solution is:

$$V_C(t) = V_0 \cdot e^{-\frac{t}{RC}}, \quad \text{for } t \geq 0$$

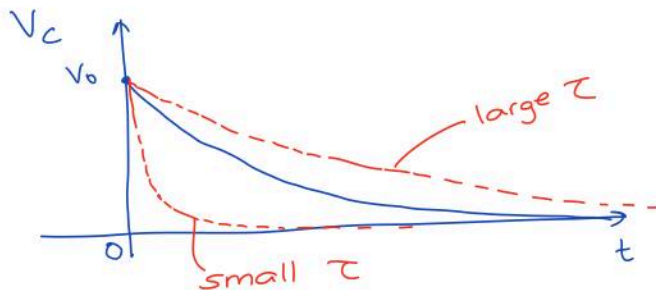
Verify the solution:

$$\frac{dV_C}{dt} + \frac{1}{RC} V_C = 0$$

$$\frac{d}{dt} (V_0 e^{-\frac{t}{RC}}) + \frac{1}{RC} V_0 e^{-\frac{t}{RC}} = 0$$

$$-\frac{1}{RC} V_0 e^{-\frac{t}{RC}} + \frac{1}{RC} V_0 e^{-\frac{t}{RC}} = 0 \quad \checkmark$$

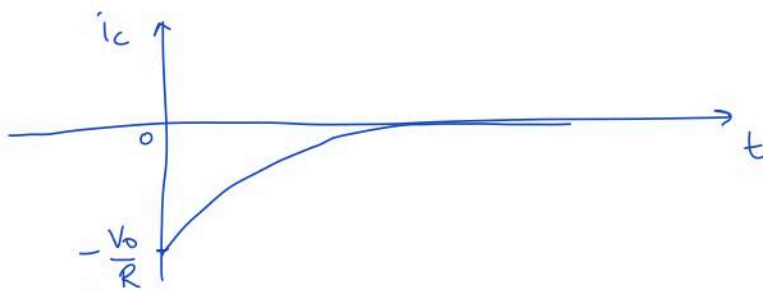
* Here, we call $\tau = RC$ is the time constant
 $1 \text{ s} = 1 \Omega \times 1 \text{ F}$



* current on the capacitor:

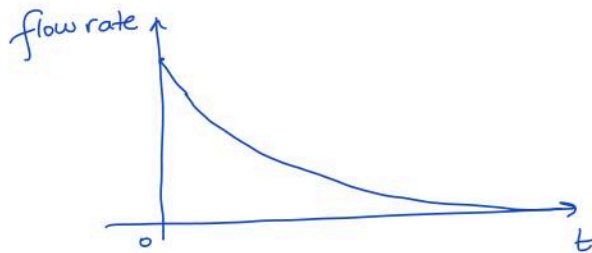
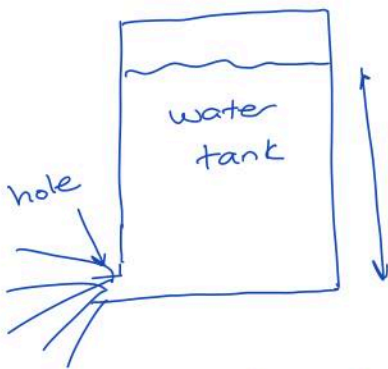
$$i_C = C \frac{dV_C}{dt} = C \frac{d}{dt} (V_0 e^{-\frac{t}{RC}}) = -C \cdot \frac{1}{RC} V_0 e^{-\frac{t}{RC}}$$

$$i_C = -\frac{V_0}{R} e^{-\frac{t}{RC}}, \quad \text{for } t \geq 0$$



negative sign means current is flowing in the opposite direction

RC time constant analogy



water $\longleftrightarrow$ charge
 flow rate $\longleftrightarrow$ current
 size of the tank $\longleftrightarrow$ capacitance
 pressure of water $\longleftrightarrow$ voltage
 size of the hole $\longleftrightarrow$ resistance
 (smaller hole) (bigger resistance)

So, small hole & big tank leads to slower decay of flow rate

Solution of first-order circuits:

(e.g., RC circuits containing just one capacitor)

- 1) Replace the capacitor with a voltage source of value equal to its initial value. $\Rightarrow V_C$ is a continuous function of time.
- 2) Solve the circuit using node analysis.
This will give V_{init} (or i_{init}).
- 3) Replace the capacitor with open circuit. $\Rightarrow$ as $t \rightarrow \infty$, capacitor acts like open circuit.
- 4) Solve the circuit using node analysis.
This will give V_{final} (or i_{final}).
- 5) Kill all sources in the circuit.
current sources set to $I=0 \Rightarrow$ open circuit
voltage sources set to $V=0 \Rightarrow$ short circuit

Find the total resistance connected in parallel with C.

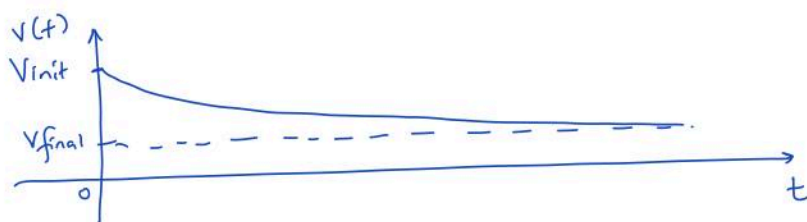
Then, $\tau = R_T \cdot C$

- 6) Write the solution:

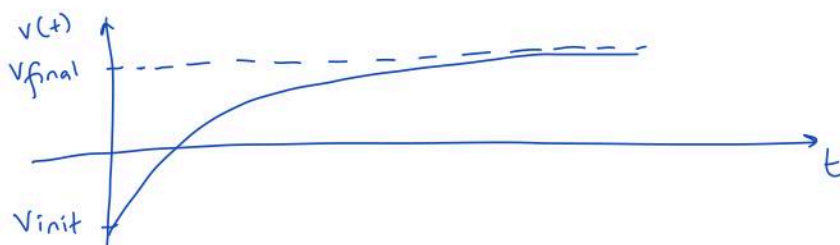
$$V(t) = V_{final} + (V_{init} - V_{final})e^{-t/\tau}, \quad t \geq 0$$

check $V(0) = V_{final} + V_{init} - V_{final} = V_{init} \quad \checkmark$

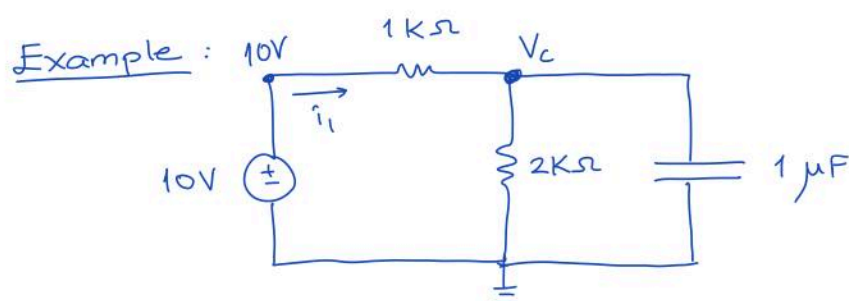
$V(t) \big|_{t \rightarrow \infty} = V_{final} \quad \checkmark$



if $V_{init} > V_{final}$



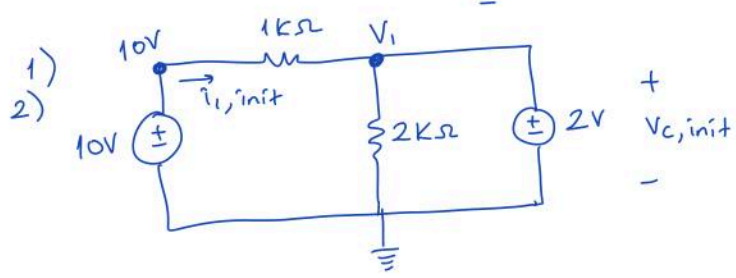
if $V_{final} > V_{init}$



$$V_c(0) = 2V$$

$$V_c(t) = ? \text{ for } t \geq 0$$

$$i_1(t) = ?$$

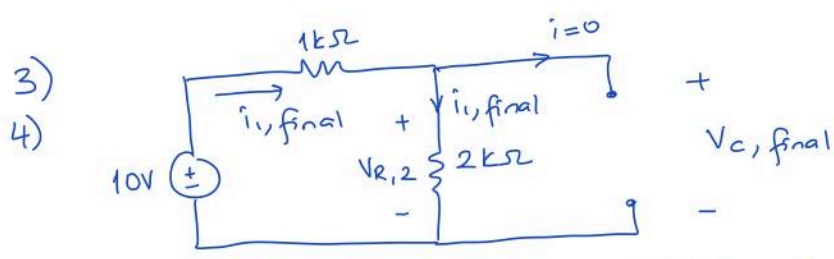


effective circuit at $t=0$

$$V_{c, \text{init}} = 2V$$

$$V_i = V_{c, \text{init}} = 2V$$

$$i_{1, \text{init}} = \frac{10 - V_i}{1k\Omega} = 8 \text{ mA}$$



effective circuit as $t \rightarrow \infty$

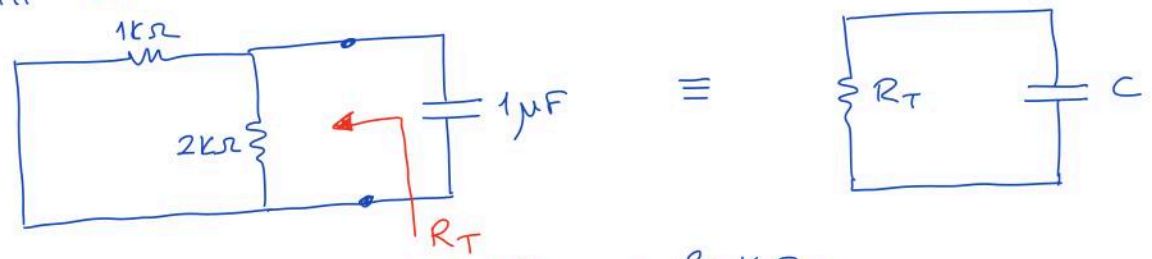
$$i_c = C \frac{dV_c}{dt}$$

$\rightarrow 0$ capacitor acts like open circuit

$$i_{1, \text{final}} = \frac{10V}{1K + 2K} \approx 3.33 \text{ mA}$$

$$V_{R,2} = i_{1, \text{final}} \times 2k\Omega = 6.67 \text{ V} = V_{c, \text{final}}$$

5) Kill all sources:



$$R_T = 1K // 2K = \frac{1K \cdot 2K}{1K + 2K} = \frac{2}{3} K\Omega$$

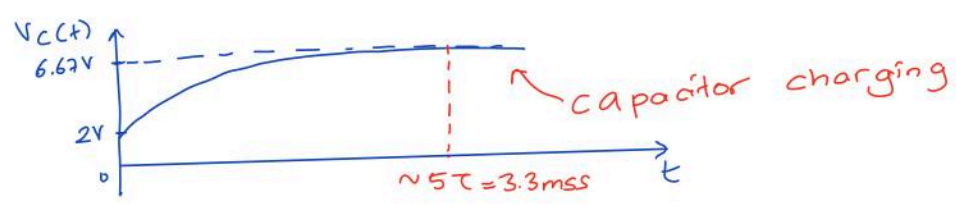
$$\tau = R_T \cdot C = \frac{2}{3} K\Omega \times 1\mu F = \frac{2}{3} \text{ ms}$$

$\underbrace{\hspace{1cm}}_{10^3} \quad \underbrace{\hspace{1cm}}_{10^{-6}}$

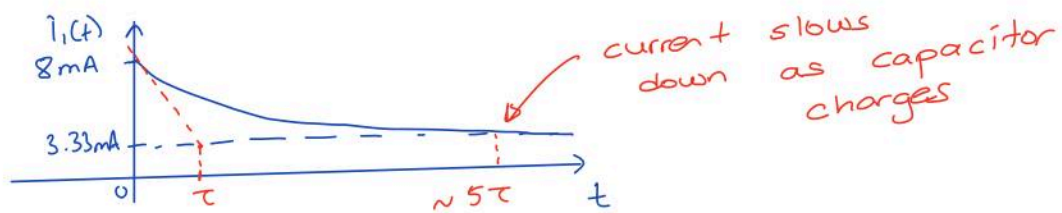
$$6) \quad V_c(t) = V_{\text{final}} + (V_{\text{init}} - V_{\text{final}}) e^{-t/\tau}$$

$$= 6.67 + (2 - 6.67) e^{-t/\tau}$$

$$= 6.67 - 4.67 e^{-t/\tau} \text{ V}, \quad t \geq 0$$



$$\begin{aligned}
 i_1(t) &= i_{1, \text{final}} + (i_{1, \text{init}} - i_{1, \text{final}}) e^{-t/\tau} \\
 &= 3.33 + (8 - 3.33) e^{-t/\tau} \\
 &= 3.33 + 4.67 e^{-t/\tau} \text{ mA}, \quad t \geq 0
 \end{aligned}$$



* at $t = 5\tau$

$$V_c(5\tau) = V_{\text{final}} + (V_{\text{init}} - V_{\text{final}}) \underbrace{e^{-\frac{5\tau}{\tau}}}_{e^{-5} \leftarrow \text{very small number} \approx 0}$$

$$V_c(5\tau) \approx V_{\text{final}}$$

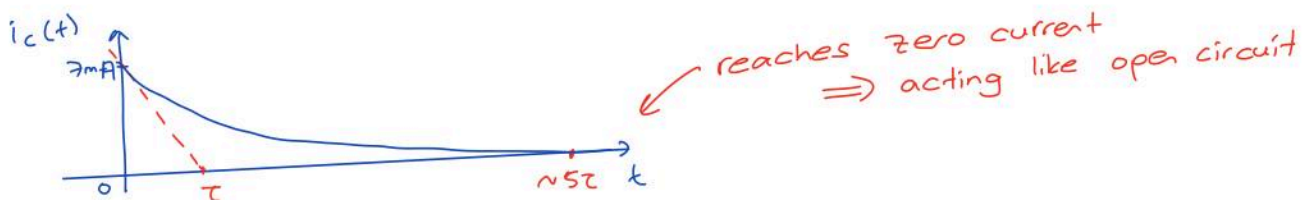
* Also,

$$\begin{aligned}
 i_1(t) &= \frac{10 \text{ V} - V_c}{1 \text{ k}\Omega} = \frac{10 - (6.67 - 4.67 e^{-t/\tau})}{1 \text{ k}\Omega} \\
 &= 3.33 + 4.67 e^{-t/\tau} \text{ mA}, \quad t \geq 0
 \end{aligned}$$

✓ same as before

* Also,

$$\begin{aligned}
 i_c(t) &= C \frac{dV_c}{dt} = 10^{-6} \frac{d}{dt} (6.67 - 4.67 e^{-t/\tau}) \\
 &= 10^{-6} \cdot \frac{4.67}{\tau} e^{-t/\tau} = 10^{-6} \cdot \frac{4.67}{\frac{2}{3} \cdot 10^{-3}} e^{-t/\tau} \\
 &\approx 7 e^{-t/\tau} \text{ mA}, \quad t \geq 0
 \end{aligned}$$



Side Notes for Capacitor :

$$i(t) = C \frac{dV(t)}{dt}$$

$$E = \frac{1}{2} CV^2$$

* Once capacitor voltage stabilizes,

$$i(t) = C \underbrace{\frac{dV(t)}{dt}}_{\rightarrow 0} \rightarrow 0$$

$\Rightarrow$ capacitor acts like open circuit

* Solution of 1st - order circuits with piecewise DC input:

$$V(t) = V_{\text{final}} + (V_{\text{init}} - V_{\text{final}}) e^{-t/\tau}$$

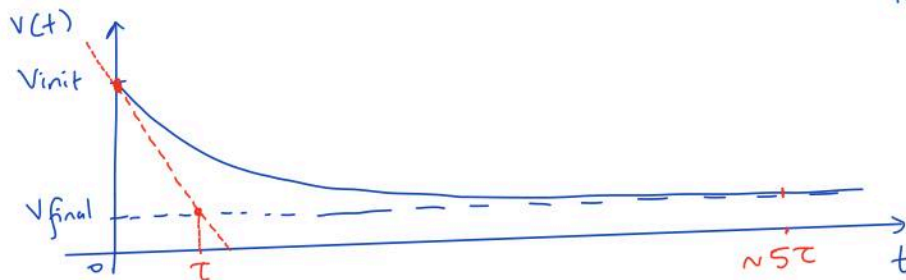
$\uparrow$ initial voltage $\uparrow$ final voltage $\leftarrow$ time constant, $\tau = R_T \cdot C$

$$V(0) = V_{\text{init}}$$

$$V(\infty) = V_{\text{final}}$$

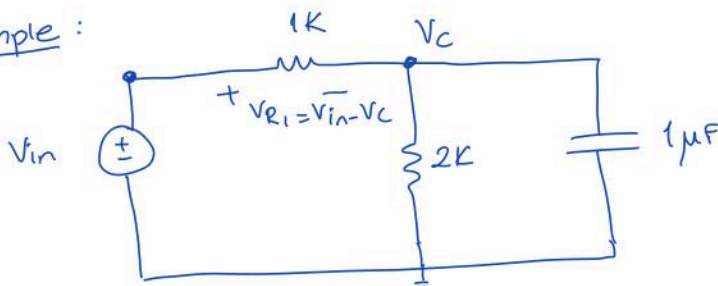
τ : decay rate

in $5\tau \rightarrow 99\%$ complete

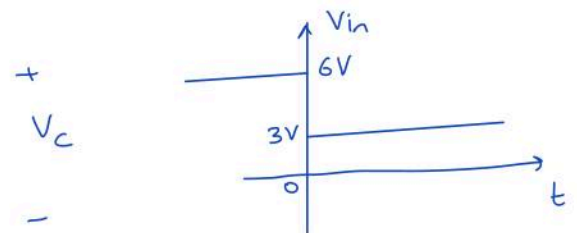


initial slope : $\left. \frac{dV}{dt} \right|_{t=0} = -\frac{1}{\tau} (V_{\text{init}} - V_{\text{final}})$

Example :

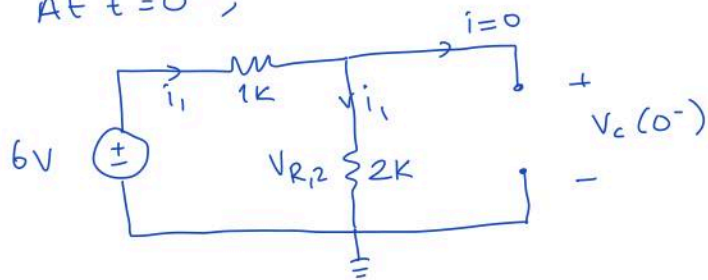


Find $V_c(t)$ for all t .



* $V_{in} = 6V$ for a very long time before $t = 0$.
Capacitor voltage must have stabilized $\rightarrow$ capacitor acts like open circuit at $t = 0^-$.

At $t = 0^-$,

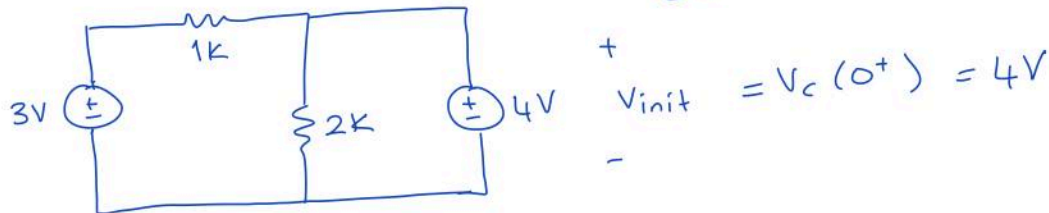


$$i_1 = \frac{6V}{1k + 2k} = 2mA$$

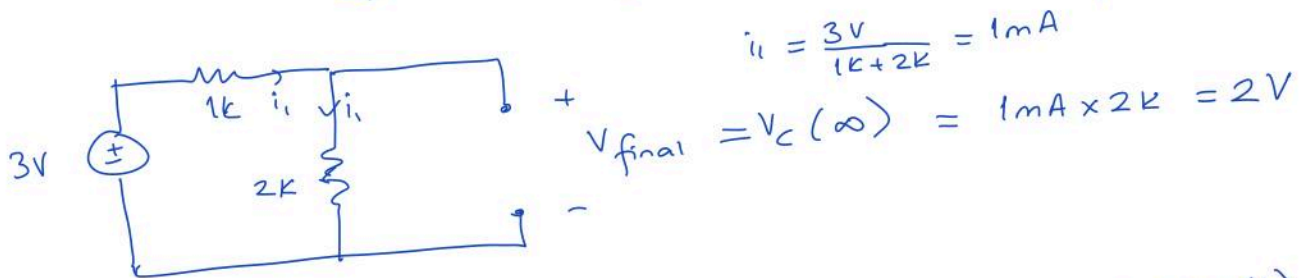
$$V_{R2} = i_1 \cdot 2k = 4V = V_C(0^-)$$

* At $t = 0^+$, capacitor voltage cannot change instantaneously.
 V_C remains at 4V momentarily.

$$V_C(0^+) = V_C(0^-) = 4V$$



* As $t \rightarrow \infty$, capacitor adjust to $V_{in} = 3V$
 $\Rightarrow$ its voltage stabilizes $\Rightarrow$ capacitor acts like open circuit.



$$i_1 = \frac{3V}{1k + 2k} = 1mA$$

$$V_{final} = V_C(\infty) = 1mA \times 2k = 2V$$

* $\tau = \frac{2}{3}ms$ as before (set V_{in} to short circuit)

* So,

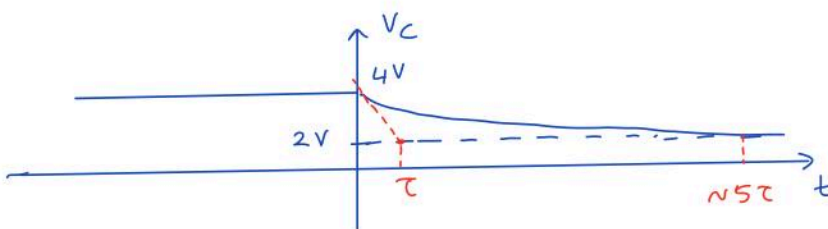
$$V_C(t) = V_{final} + (V_{init} - V_{final})e^{-t/\tau}, \quad t \geq 0$$

$$= 2 + (4 - 2)e^{-t/\tau}$$

$$= 2 + 2e^{-t/\tau} \quad V, \quad t \geq 0$$

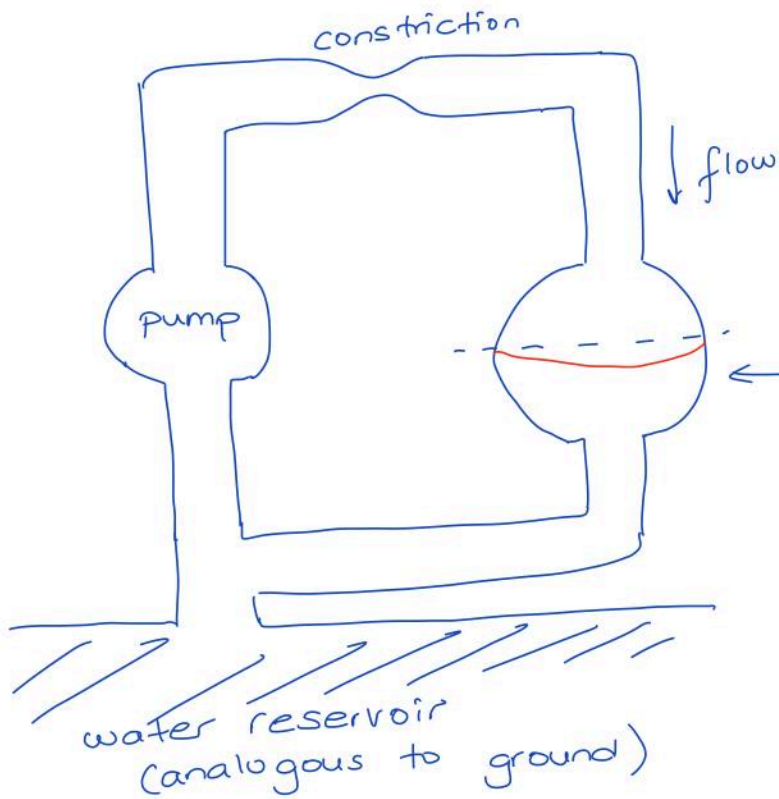
Then,

$$V_C(t) = \begin{cases} 4V & , t < 0 \\ 2 + 2e^{-t/\tau} \quad V & , t \geq 0 \end{cases}$$



Water analogy for RC circuit

29

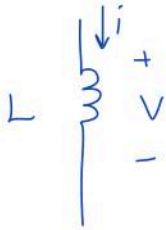


← spherical tank with
a rubber membrane
(analogous to capacitor)

membrane stretches
to fill the tank.
when stretching stops,
flow also stops.

Inductors :

- store energy in the form of magnetic flux
- resist change in current



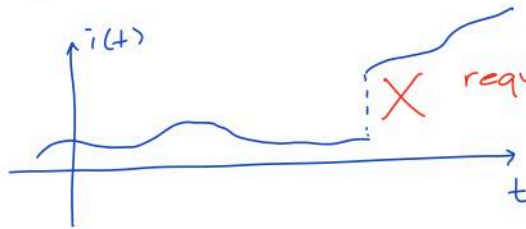
$$V = L \frac{di}{dt}$$

inductance in units of Henry (H)

- most inductors are in physical form of a wound coil (also called "solenoid")
- Inductors do not dissipate energy. They store energy.
- Net magnetic energy stored in an inductor:

$$E = \frac{1}{2} L I^2$$

* $V = L \frac{di}{dt}$

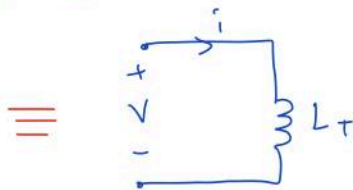
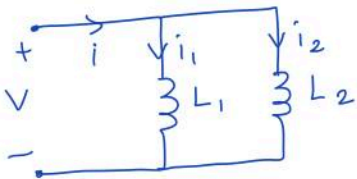


requires infinite voltage
 $\frac{di}{dt} \rightarrow \infty$
 $V \rightarrow \infty$

*** current of an inductor cannot change instantaneously!

Parallel / Series connections

- Works similar to resistors

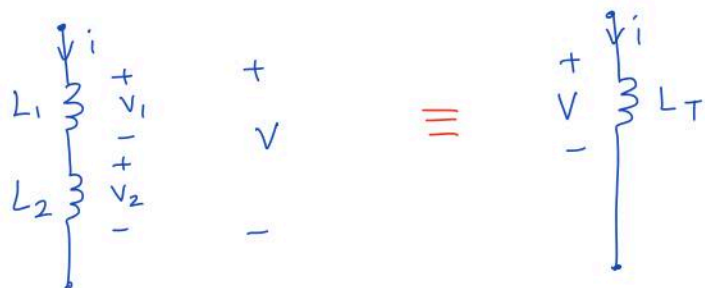


$$V = L_T \frac{di}{dt} = L_T \left(\frac{di_1}{dt} + \frac{di_2}{dt} \right)$$

$$V = L_T \left(\frac{V}{L_1} + \frac{V}{L_2} \right)$$

$$L_T = \left(\frac{1}{L_1} + \frac{1}{L_2} \right)^{-1}$$

reduced inductance

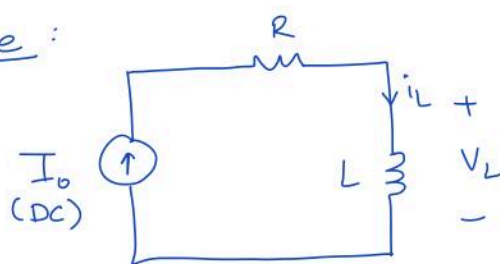


$$V = V_1 + V_2 = L_1 \frac{di}{dt} + L_2 \frac{di}{dt} = L_T \frac{di}{dt}$$

$$L_T = L_1 + L_2$$

↑ increased inductance

Example :

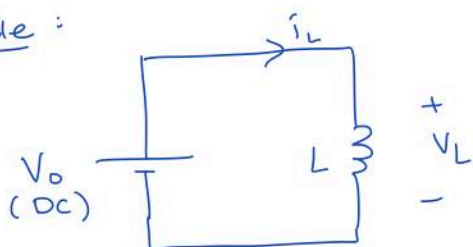


$$i_L(t) = I_0 \quad \text{constant}$$

$$V_L = L \frac{di_L}{dt} = 0$$

⇒ inductor acts like short circuit when current is constant.

Example :



what is V_L ? what is i_L ?

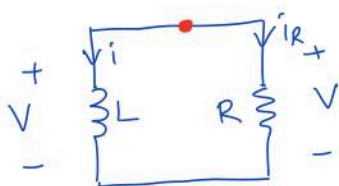
$$V_L = V_0 = L \frac{di_L}{dt}$$

$$i_L(t) = \int \frac{V_0}{L} dt$$

Assume $i_L(0) = 0$. Then,

$$i_L(t) = \frac{V_0}{L} \cdot t \Rightarrow \text{current grows indefinitely large}$$

RL Circuits :



Node analysis: $i + i_R = 0$ (KCL)

$$i + \frac{V}{R} = 0$$

$$i + L \frac{di/dt}{R} = 0$$

$$\frac{di(t)}{dt} + \frac{R}{L} i(t) = 0$$

first-order ordinary differential equation

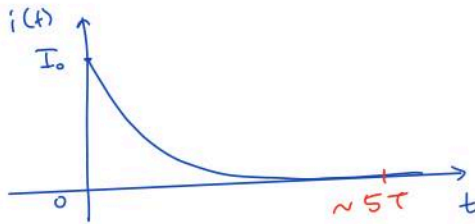
Solution : $i(t) = I_0 \cdot e^{-t/\tau}$, for $t \geq 0$

initial current

$$\tau = \frac{L}{R}$$

is the time constant

$$1s = \frac{1H}{1\Omega}$$

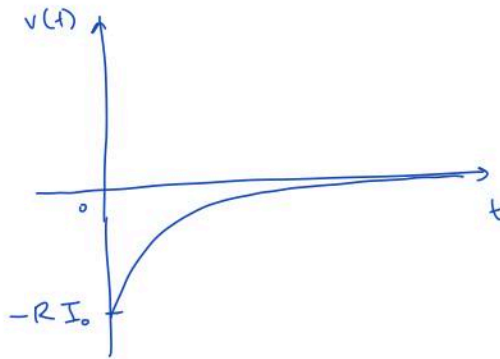


magnetic energy stored in the inductor is dissipated as heat on the resistor.

* Now,

$$v(t) = L \frac{di}{dt} = L \frac{d}{dt} (I_0 e^{-t/\tau}) = -\frac{L}{\tau} I_0 \cdot e^{-t/\tau}$$

$$= -R I_0 e^{-t/\tau}$$

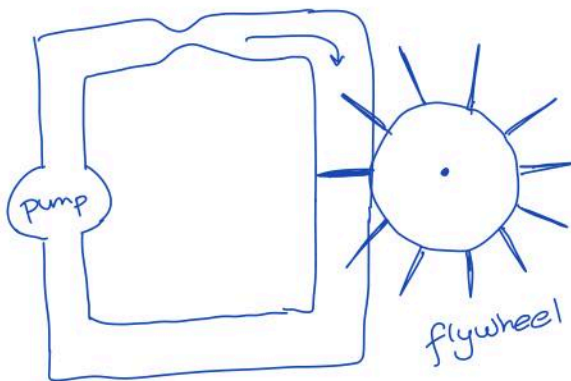


Once inductor current stabilizes,

$$V = L \frac{di}{dt} \rightarrow 0$$

* So, inductor behaves like a short circuit eventually, if the input is stable for a long time.

Water Analogy :



inductor $\longleftrightarrow$ fly wheel
current $\longleftrightarrow$ flow rate

* fly wheel initially resists water flow due to inertia

* Once the wheel starts to rotate, it will not enforce any resistance on the pipe system

Re-visit : solution of first-order circuits :

(RC circuits with just one capacitor, OR
RL circuits with just one inductor)

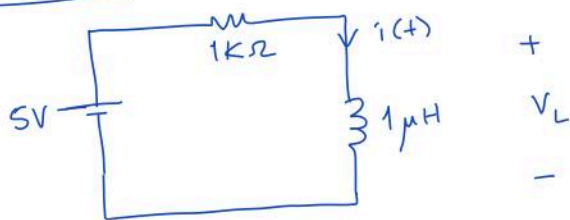
- 1) Replace the inductor with a current source of value equal to its initial value.
(capacitor is replaced by a voltage source).
- 2) Solve the circuit using nodal analysis.
This will give V_{init} , i_{init} .
- 3) Replace the inductor with a short circuit
(capacitor is replaced with an open circuit)
- 4) Solve the circuit to get V_{final} , i_{final} .
- 5) Kill all sources in the circuit:
current source (set $I=0$) $\Rightarrow$ open circuit
voltage source (set $V=0$) $\Rightarrow$ short circuit
Find the total resistance in parallel with inductor
(or capacitor).

6) Write the solution:

$$V(t) = V_{final} + (V_{init} - V_{final})e^{-t/\tau}, \quad t \geq 0$$

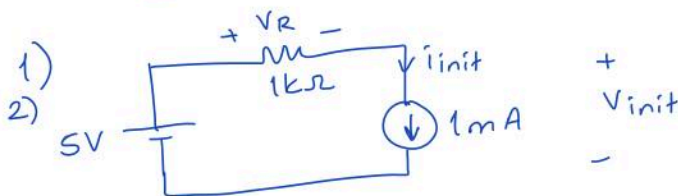
$$\text{where } \tau = \frac{L}{R} \quad \text{OR} \quad \tau = RC$$

Example :



$$i(0) = 1 \text{ mA}$$

$$i(t) = ? , \quad V_L(t) = ? \quad \text{for } t \geq 0$$

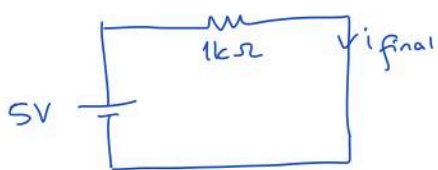


$$i_{init} = 1 \text{ mA}$$

$$V_{init} = 5V - V_R = 5 - 1\text{mA} \times 1\text{k}\Omega$$

$$= 4V$$

- 3) Eventually, inductor current will stabilize
 4) $\Rightarrow$ inductor behaves like short circuit



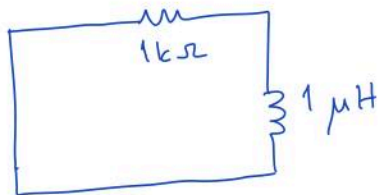
$$+ \quad V_{final}$$

$$- \quad$$

$$V_{final} = 0$$

$$i_{final} = \frac{5V}{1k\Omega} = 5mA$$

- 5) Kill all sources:



$$R_T = 1k\Omega$$

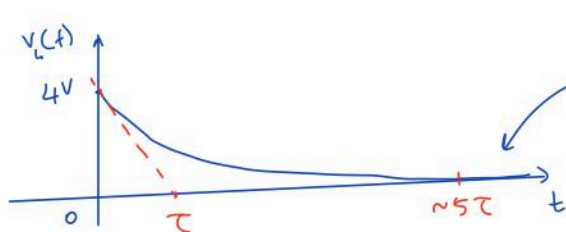
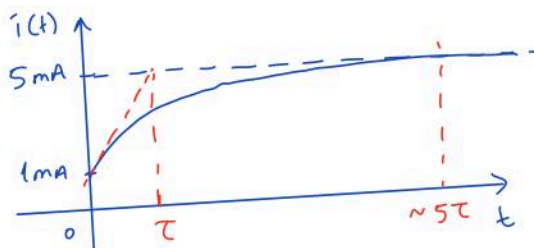
$$\tau = \frac{L}{R} = \frac{1\mu H}{1k\Omega} = 1ns$$

$$6) \quad i(t) = i_{final} + (i_{init} - i_{final})e^{-t/\tau}$$

$$= 5 + (1 - 5)e^{-t/\tau} = 5 - 4e^{-t/\tau} \text{ mA}, t \geq 0$$

$$V_L(t) = V_{final} + (V_{init} - V_{final})e^{-t/\tau}$$

$$= 0 + (4 - 0)e^{-t/\tau} = 4e^{-t/\tau} \text{ V}, t \geq 0$$

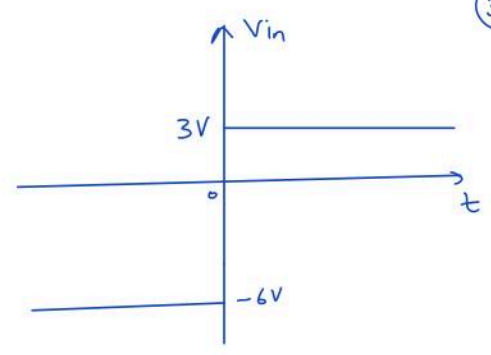
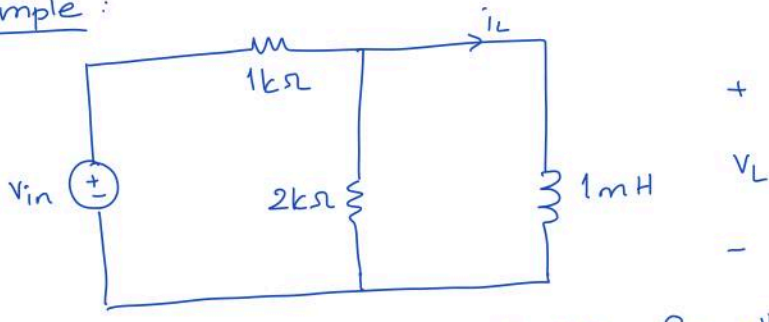


inductor eventually behaves like a short circuit

* Double check:

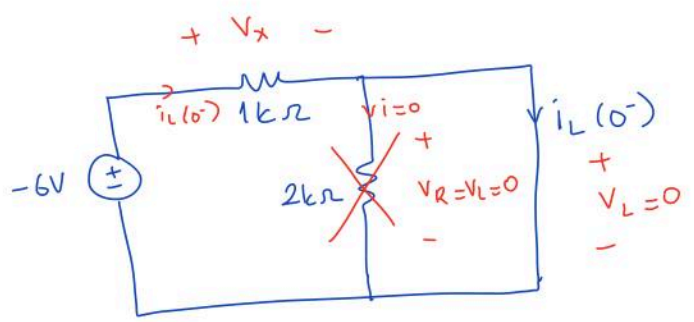
$$V_L(t) = L \frac{di}{dt} = 4e^{-t/\tau} \text{ V} \quad (\text{exercise})$$

Example :



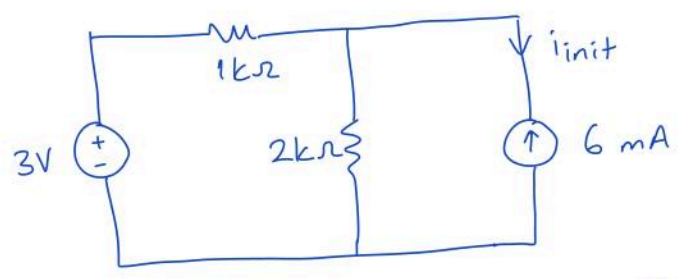
Find $i_L(t)$ and $V_L(t)$ for all t .

* $V_{in} = -6V$ for an infinitely long time before $t=0$.
Effective circuit at $t=0^-$: inductor must have adjusted to this input voltage $\Rightarrow$ behaves like short circuit



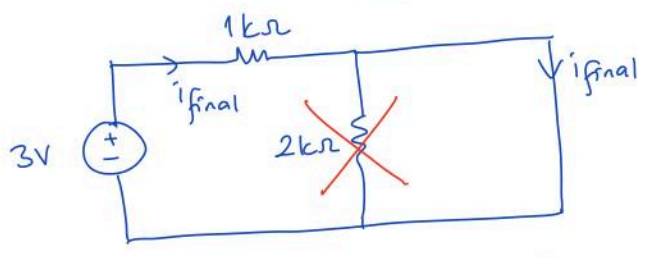
$$V_x = -6V$$
$$i_L(0^-) = \frac{-6V}{1k\Omega} = -6mA$$

* At $t=0^+$, inductor current cannot change instantaneously.
so, $i_L(0^+) = i_L(0^-) = -6mA = i_L(0)$



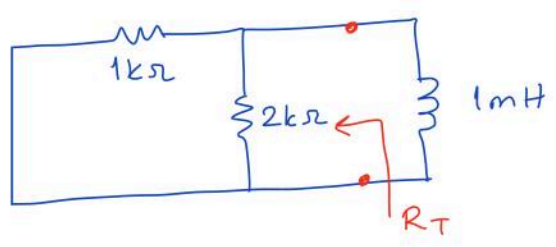
$$i_{init} = -6mA$$

* As $t \rightarrow \infty$, inductor will adjust to $V_{in} = 3V$ and behave like short circuit



$$i_{final} = \frac{3V}{1k\Omega} = 3mA$$

* τ : kill all sources

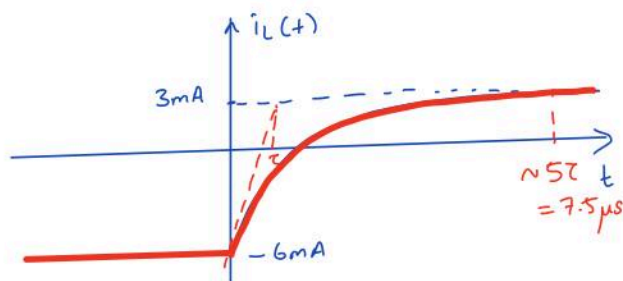


$$R_T = 1k\Omega // 2k\Omega = \frac{1 \cdot 2}{1+2} k\Omega = \frac{2}{3} k\Omega$$

$$\tau = \frac{L}{R_T} = \frac{1mH}{\frac{2}{3}k\Omega} = 1.5 \mu s$$

* So,

$$\begin{aligned}
 i_L(t) &= i_{\text{final}} + (i_{\text{init}} - i_{\text{final}}) e^{-t/\tau} \\
 &= 3 + (-6 - 3) e^{-t/\tau} \\
 &= 3 - 9e^{-t/\tau} \text{ mA}, \quad t \geq 0
 \end{aligned}$$



$$i_L(t) = \begin{cases} -6 \text{ mA}, & t < 0 \\ 3 - 9e^{-t/\tau} \text{ mA}, & t \geq 0 \end{cases}$$

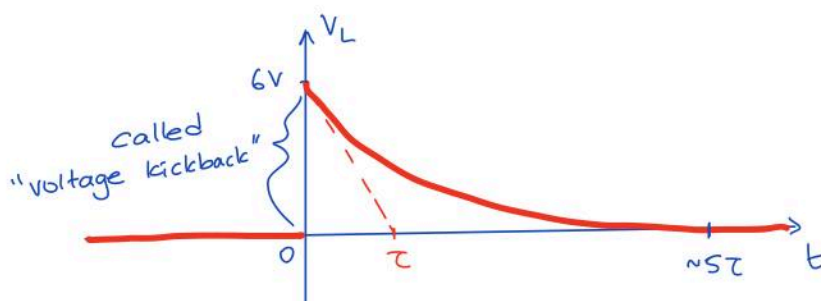
* Then, $v_L(t) = 0$, $t < 0$

$$\text{for } t \geq 0, \quad v_L(t) = L \frac{di_L}{dt} = 10^{-3} \cdot \frac{d}{dt} (3 - 9e^{-t/\tau}) \cdot 10^{-3}$$

$$= 10^{-6} \cdot \frac{9}{\tau} e^{-t/\tau} = \frac{10^{-6} \cdot 9}{1.5 \times 10^{-4}} e^{-t/\tau}$$

$$= 6e^{-t/\tau} \text{ V}, \quad t \geq 0$$

$$v_L(t) = \begin{cases} 0, & t < 0 \\ 6e^{-t/\tau} \text{ V}, & t \geq 0 \end{cases}$$

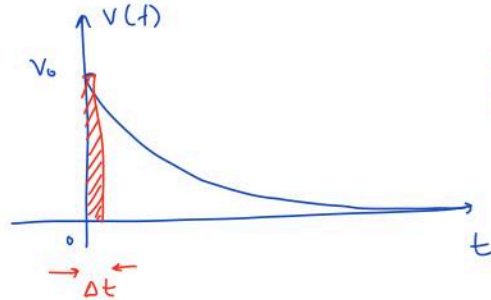
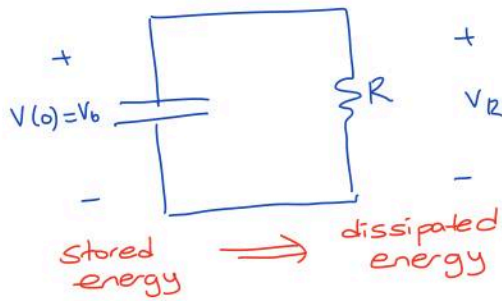


Time constant : $\tau = \underline{RC}$

, $\tau = \underline{\frac{L}{R}}$. Why?

(37)

* For RC case :



initial stored energy $= \frac{1}{2} C V_0^2$

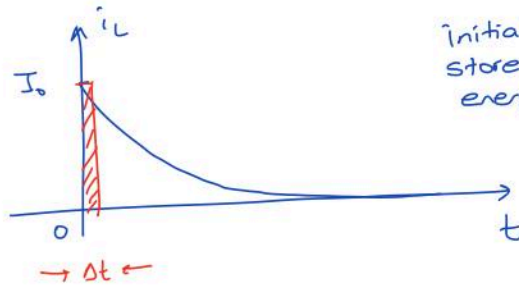
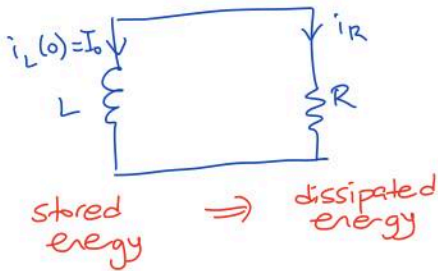
For Δt interval : $V_R = V_0$

$$P = I \cdot V = \frac{V^2}{R} = \frac{V_0^2}{R}$$

$\Rightarrow$ less energy dissipated per unit time if R is big.

$\Rightarrow$ long time constant : $\tau = \underline{RC}$

* For RL case :



initial stored energy $= \frac{1}{2} L I_0^2$

* For Δt interval : $i_R = -I_0$

$$P = I \cdot V = I^2 R = I_0^2 R$$

$\Rightarrow$ more energy dissipated per unit time if R is big

$\Rightarrow$ short time constant : $\tau = \underline{\frac{L}{R}}$

* Similar to :

Potential Energy Release

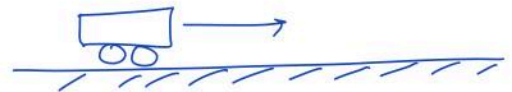


(like RC circuit)

friction $\longleftrightarrow$ resistance
more friction $\longleftrightarrow$ takes longer to release potential energy

Kinetic Energy Release

(like RL circuit)



more friction $\longleftrightarrow$ quicker dissipation of kinetic energy